Adaptive Digital Wellbeing Coach

Literature, Data and Technology Review

Samsung Innovation Campus Capstone Project | Group 8

Literature Review

1.1 Introduction

Our project aims to develop a personalized Android application that anticipates potentially unwanted smartphone use and provides brief, context-sensitive reminders. The intended users are adults aged 18 and above who want greater control over their smartphone habits. The system will support awareness and self-regulation, it will not diagnose or treat smartphone addiction.

We focus on usage that conflicts with a person's intentions or goals. Long sessions, repeated checking and late-night activity may help identify relevant patterns, but none of these behaviours is necessarily problematic on its own. Understanding this distinction is essential when defining model targets and deciding when to notify users.

The literature review examines three questions: how unwanted use can be measured, what machine learning can predict from available data, and how personalization and feedback can support interventions. These questions inform our proposed comparison of a general model, a personalized model and a model trained only on an individual user's data.

1.2 Measuring Smartphone Use and Its Meaning

Marciano and Camerini (2022) studied 84 adolescents in Switzerland using 45 days of smartphone tracking and questionnaire measurements collected one year apart. They examined usage duration, frequency and the difference between recorded and self-reported duration. Recorded duration was associated with concurrent problematic smartphone use, whereas overestimating duration predicted later scores after controlling for baseline scores. Frequency was not significant in the main models. The small sample and narrow age range limit generalization, but the study demonstrates why recorded behaviour and subjective evaluation should be considered separately. It does not establish a universal screen-time threshold for problematic use.

Parry and Toth (2025) provide a methodological guide for extracting meaningful measures from Android event logs. Their procedure distinguishes glances, phone sessions and application episodes, addressing the difficulty of reconstructing behaviour from individual system events. The supplied review notes describe an illustration using 5,000 events from 10 devices. This is a measurement study rather than a risk-prediction experiment. Its main contribution to our project is a reproducible approach to session reconstruction and feature extraction. However, the procedure must be adapted to the event types actually available in our dataset and application.

Guo et al. (2025) followed 17 Android users for one week, combining screenshots, experience sampling, surveys and interviews. Their social media analysis covered 34,313 screenshots. A multimodal language model helped identify activities, while participants reported their regret. Non-intentional use and viewing algorithmically recommended content or comments were associated with greater regret. The small sample and privacy demands of screenshot collection limit direct adoption. For our project, the relevant finding is that usage duration alone cannot capture whether an experience

was unwanted. We will use brief feedback rather than collect screenshots or attempt to infer screen content.

1.3 Machine Learning for Risk and Engagement Prediction

Lee and Kim (2021) compared Decision Tree, Random Forest and XGBoost using a South Korean survey. The source survey contained 29,712 respondents, the selected age groups used for modelling totalled 14,192 participants. The models used 27 questionnaire-derived variables, and Random Forest achieved a reported average accuracy of 82.59%. Although the paper discusses future applications to phone logs, its experiment used survey responses. It supports comparing tree-based models, but does not validate real-time prediction from passive logs or provide labels for our dataset.

Kim, Yoon and Shin (2024) analysed survey data from 7,379 South Korean children and adolescents aged 10–19. They compared eight machine learning and deep learning algorithms and used SHAP to examine model predictions. XGBoost achieved a reported F1 score of approximately 0.825 and precision of 0.876. These results concern a questionnaire-defined risk group, rather than individual unwanted-use episodes. The study supports evaluating tree-based models and feature explanations, but its performance cannot be transferred directly to our adult population, behavioural logs or prediction target.

Pielot et al. (2017) investigated when users would engage with suggested content after receiving a notification. Their analysis included 337 Android users and more than 120 million usage events. XGBoost performance improved when previous responses were included: F1 increased from 0.113 to 0.311. The frequently cited fivefold improvement refers to precision relative to a simple baseline, not overall accuracy. This study informs our feature engineering and participant-based evaluation. Its engagement labels do not represent overuse, unwanted use or willingness to stop using the phone.

1.4 Personalized Interventions and User Feedback

Monge Roffarello and De Russis (2021) developed a method for discovering habitual smartphone use through clustering and association-rule mining. They evaluated the method using data from 35 people and subsequently assessed their Socialize application with 20 students. Users selected habits they wanted to avoid and supplied alternative activities. The study reported a reduction in time associated with targeted habits, but lacked a control group. Its contribution is the distinction between discovering a recurring pattern and deciding that the pattern is unwanted: the latter remained the user's choice.

Orzikulova et al. (2024) developed Time2Stop, an adaptive intervention system evaluated in an eight-week study with 71 Android users. It combined usage and contextual features with user-provided overuse labels, updated personalized models and offered SHAP-based explanations. Its initial Random Forest model achieved an F1 score of 0.667. Adaptation improved intervention-related outcomes, although effects differed across measures; the reduction in usage duration was not statistically significant. Time2Stop provides a direct reference for feedback-based adaptation and highlights the difficulty of collecting sufficient labels when a new user joins.

1.5 Synthesis and Research Gap

The studies address related but different targets: questionnaire-defined risk, habitual behaviour, regret, engagement and momentary overuse. Their performance measures should therefore not be treated as

directly comparable. Survey studies offer larger samples but rely on self-report, field studies provide richer behavioural context but often involve smaller, selective populations.

Together, the findings support combining behavioural records with user evaluation. Passive logs describe what happened, while feedback helps establish whether it conflicted with the user's intentions. Neither source removes the limitations of the other.

Our project will examine a specific question: **does information learned from multiple users improve prediction during the early stages of personalization, and how does this benefit change as personal data accumulates?**

This is a focused evaluation objective rather than a claim that adaptive smartphone interventions are new. A central challenge is that our public dataset lacks unwanted-use labels. We will distinguish experiments using behavioural proxy labels from later evaluation using user feedback. Improvements on proxy labels alone will not be presented as evidence that the system recognizes genuinely unwanted use or reduces addiction.

Data Research

2.1 Research Objectives

The data work will investigate:

1. Whether recent usage patterns can predict elevated usage in a subsequent time window.
2. Whether personalization improves performance over a fixed general model.
3. How performance changes with the amount of personal history and labelled feedback.
4. Whether behavioural proxy predictions agree with users' own evaluations.

At this stage, we have reviewed the dataset documentation and relevant literature. Full preprocessing, exploratory analysis and model training remain planned work.

2.2 Planned Data Sources

Source	Description	Intended role	Main limitation
Mobile Phone Use Dataset	Logs from 342 participants, collected over several weeks in summer 2016; approximately 310 MB compressed, with one CSV file per participant	Develop preprocessing, extract common features and evaluate initial models	No direct unwanted-use labels; historical devices and usage patterns
Consenting users of our Android prototype	Planned collection of usage events and brief evaluations of recent use	Assess transfer to current devices and personalize predictions	Recruitment, incomplete feedback and limited observation time

The public dataset contains events from 25 physical and virtual sensors, including foreground applications and notifications, alongside emotion reports and some participant traits. Its release contains 342 participants, whereas the associated publication analysed 337. These participant counts

should be distinguished when describing the dataset and the study. The provider also notes that the original analysis scripts and non-anonymized data are unavailable.

We selected this dataset because repeated observations from multiple users suit our comparison of general and personalized models. However, the presence and completeness of each required field must be checked before finalizing the feature set. We will also verify the applicable reuse conditions before redistributing any data, public download availability does not establish unrestricted reuse.

2.3 Preprocessing and Feature Extraction

We will first inspect timestamps, event definitions, recording coverage, duplicate records and missing event sequences. Phone sessions and application episodes will be reconstructed where the available events support them. Recording gaps will be flagged rather than interpreted as periods of no phone use.

The initial modelling unit will be a 30-minute window. Candidate features include:

Feature group	**Examples**
Usage intensity	Screen-active duration, session count, application changes
Session characteristics	Previous session duration, recent average session duration
Checking behaviour	Unlock count and time between unlocks, where reliably observable
Temporal context	Hour of day, weekday and activity during a defined night period
Personal history	Deviation from the user's previous usage patterns
Previous feedback	Historical unwanted-use rate and earlier reminder responses, when available

Personal averages and feedback summaries will use only information available before prediction. Features that cannot be reproduced reliably in both the public dataset and the Android prototype will be excluded from the transferable model.

2.4 Labels and Prediction Design

We will maintain two distinct label sources.

Behavioural proxy labels. For initial experiments, we will define elevated-use patterns using indicators such as unusually long sessions, repeated checking and concentrated night-time activity. Thresholds will be documented, selected using training data and tested for sensitivity. These labels describe our

operational definition of elevated usage; they are not validated measures of addiction or unwanted use.

To evaluate anticipation, features from a completed window will predict the proxy outcome in the following window. Using a window's screen time both to define its label and to predict that same label would mainly reproduce the labelling rule.

User feedback labels. In the prototype, a short question will ask whether a specified recent period of use continued longer than the user wanted. Responses will be stored with the relevant time interval and feature record. An explicit negative response will be distinguished from a skipped question.

The response will serve as a target label, not an input to the prediction it evaluates. Earlier responses may inform later predictions. Whether a user accepts or dismisses a reminder will be stored separately, because dismissing an intervention does not establish that the preceding use was wanted.

Proxy and feedback labels will retain separate source identifiers. Their agreement will be assessed before combining them in any training procedure.

2.5 Planned Analysis and Evaluation

The exploratory analysis will report recording days per participant, valid-window coverage, screen-active duration, session distributions, missingness and the prevalence of each proxy label. Planned visualizations include daily usage distributions, hourly activity profiles and variation between users.

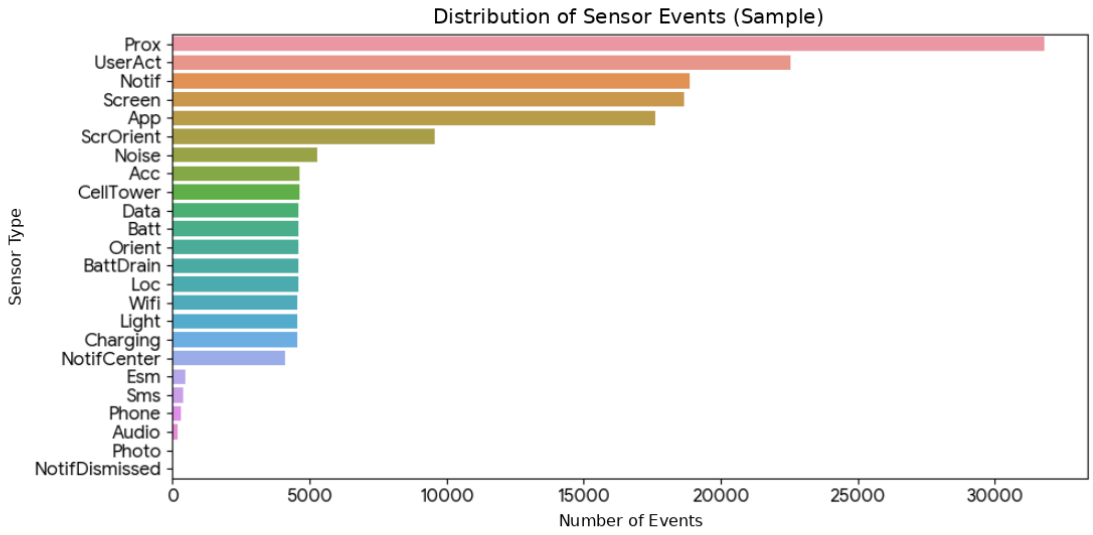


Figure 1. Distribution of Sensor Events

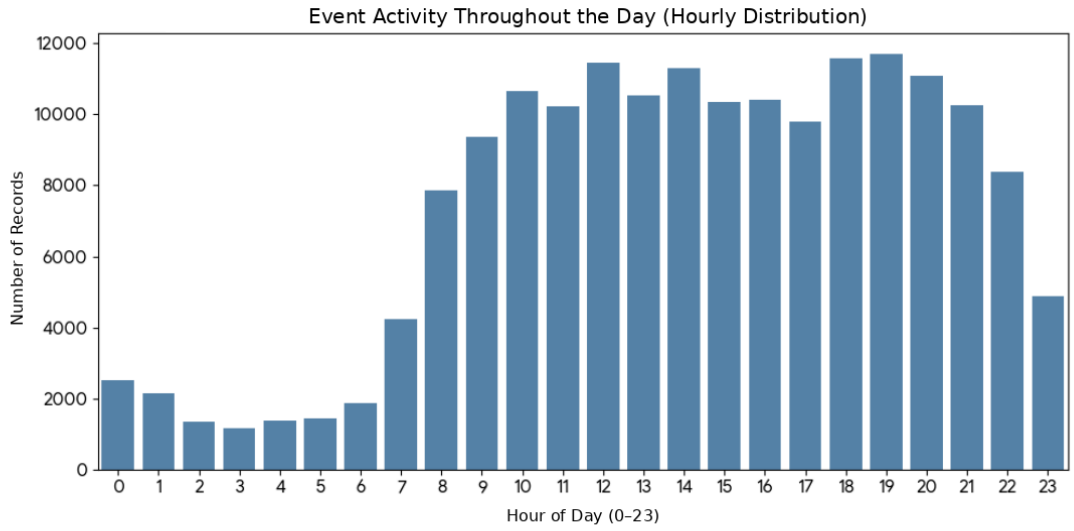


Figure 2. Event Activity Throughout the Day (Hourly Distribution)

We do not yet report these as observed results. In particular, sensor-event counts will not be interpreted directly as screen time, since event types have different recording frequencies.

The model comparison will include:

- **General model:** trained on public-data participants, without adaptation to the evaluation user.
- **Personalized model:** combines population information with the evaluation user's earlier records.
- **Personal-only model:** trained exclusively on that user's earlier records.

We will first compare these approaches using the same proxy target. A later evaluation will assess predictions against independently collected user feedback. This second stage will test whether transfer remains useful when the target changes from a behavioural rule to subjective unwanted use.

Evaluation participants will be excluded from general-model training. Personalization will use earlier records, with testing on later records and no overlapping feature or outcome intervals across the

boundary. Where data permits, we will compare several history lengths, such as 1, 3, 7 and 14 days, and report the number of available labels alongside elapsed time.

Metrics will include precision, recall, F1 and precision–recall AUC, together with per-user results and false alerts per day. Any class weighting or resampling will be restricted to training data. If feedback is too sparse, we will report feasibility findings rather than make unsupported claims about personalized performance.

2.6 Ethics and Data Privacy

The initial pilot will involve adults only. Before collection, participants will receive an explanation of the data collected, its purpose, storage arrangements and withdrawal options. We will obtain informed consent and address applicable institutional requirements before recruitment.

Collection will be limited to usage metadata needed for the research. Screenshots, message contents, contact lists and precise location are outside the initial scope. Participant identifiers will be pseudonymous; replacing names with codes alone will not be described as complete anonymization.

We plan to retain raw logs locally where practical. Any research export will be separately explained and limited to necessary records, with restricted access and secure storage. Retention periods and deletion procedures will be specified before collection. These measures will be designed around KVKK requirements concerning purpose limitation, proportionality, lawful processing and retention. On-device processing alone will not be treated as proof of compliance.

2.7 Conclusion

The public dataset is a suitable candidate for developing the analysis pipeline and testing generalization across users. Its main limitation is the absence of a target that directly represents unwanted use. Our data plan therefore separates reproducible proxy experiments from user-feedback validation and treats transfer to current Android users as an empirical question.

Technology Review

3.1 Technical Scope

The proposed system requires usage-data collection, feature extraction, prediction, personalization and a lightweight feedback interface. Technology choices will prioritize reproducibility, limited data collection and manageable development effort. The first implementation will focus on a small set of reliable features rather than extensive sensing.

3.2 Android Data Collection

Android's **UsageStatsManager** will be the starting point for collecting usage history. Most relevant operations require usage access granted by the user through Settings. Event availability, retention and behaviour across supported Android versions must be tested; the API should not be assumed to provide every feature used in the reviewed studies.

We propose a native Android prototype using Kotlin. An early feasibility test will check event reconstruction, background operation and battery impact before integrating a prediction model. Scrolling dynamics and screenshot analysis are outside the initial scope.

3.3 Model Comparison

Approach	Strengths	Limitations	Planned role
Fixed or user-adjustable rules	Transparent, inexpensive and easy to deploy	Limited ability to represent interactions between features	Practical baseline
Logistic Regression	Lightweight and straightforward to evaluate	Requires suitable feature design for complex relationships	Statistical baseline
Random Forest	Captures nonlinear relationships in tabular data	Model size and adaptation procedure require attention	Main candidate
XGBoost	Flexible tabular modelling with regularization and weighting	Requires tuning and deployment validation	Main candidate
Sequence models such as LSTM	Can represent temporal sequences directly	Greater implementation complexity and data demands	Possible extension

Random Forest and XGBoost are justified candidates, not guaranteed winners. Results from questionnaire studies do not establish their performance on our logs. Python-based preprocessing and modelling will allow the alternatives to be evaluated on identical splits and targets.

For the first personalization experiment, we propose retraining with pooled records while assigning greater weight to the current user's earlier data. This gives both tree-based candidates an explicit, reproducible adaptation procedure. We will avoid describing this generically as neural-network-style fine-tuning.

Initial training will run in the research environment. Mobile inference and any on-device adaptation will be assessed separately for compatibility, latency, memory use and battery cost.

3.4 Explainability and Feedback

SHAP will be considered for examining which features contribute to tree-model predictions. Its TreeExplainer supports ensemble tree models, although interpretation depends on choices such as the treatment of correlated features (SHAP developers, n.d.).

User-facing explanations will describe observed behaviour in plain language. They will not claim that a feature caused harmful use or that the model has diagnosed a condition. Before presenting personal comparisons, the system must have sufficient history to support them.

Reminders will encourage reflection or a user-selected alternative activity. Users will be able to dismiss, pause or disable them. Notification limits and quiet periods will help prevent the application from becoming another source of interruption. Reminder wording will draw on self-monitoring and goal-setting principles without presenting the application as cognitive behavioural therapy.

3.5 Existing Applications and Research Examples

Google's Digital Wellbeing provides usage summaries, application timers, Focus mode and bedtime-related settings. These features establish useful practical baselines for awareness and user-controlled limits (Google, n.d.). Our evaluation will ask whether learned predictions offer additional value over such simple controls, rather than assume that existing tools lack personalization.

Time2Stop demonstrates feedback-based adaptation and explainable interventions, while Socialize demonstrates user-selected habits and alternative activities. We will draw on these design principles while keeping the prototype's sensing and interaction requirements limited.

3.6 Limitations and Conclusion

The main technical uncertainties are reliable event collection, differences between historical and current usage data, sparse feedback and the deployment of personalized models. The main interaction risk is excessive or poorly timed prompting.

We will address these issues through staged development: validate collection, establish baselines, evaluate personalization and then integrate reminders. Predictive performance, usability and behavioural outcomes will be evaluated separately. A working classifier will not by itself demonstrate reduced unwanted use.

The initial technology choices are intended to support a testable prototype with transparent assumptions. The application will provide awareness support, with information about professional support services where appropriate, rather than act as a diagnostic or therapeutic system.

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